MODULE-VI

Introduction to computational complexity

TOPICS

- Tractable and Intractable Problems
- Complexity Classes
  - P and NP Classes -
    - ✓ SAT and 3-SAT Problems
  - NP-Hard and NP-Complete Classes
  - Study of NP complete problems
    - ✓ Travelling Sales Person Problem
    - ✓ Knapsack Problem
    - ✓ Clique Problem
    - ✓ Vertex Cover Problem.

TRACTABLE AND INTRACTABLE PROBLEMS

Tractable Problem

- A problem that is solvable by a polynomial-time algorithm.
- The upper bound is polynomial.

Polynomial-Time Algorithm:

- An algorithm whose order-of-magnitude time performance is bounded from above by a polynomial function of n, where n is the size of its inputs.
- For input size n, if worst-case time complexity of an algorithm is $O(n^k)$, where k is a constant, the algorithm is a polynomial time algorithm.

Polynomial functions:

- ❖ Any function that is $O(n^k)$, i.e. bounded from above by n^k for some constant k. E.g. O(1), O(log n), O(n), O(n × log n), $O(n^2)$, $O(n^3)$

Examples

Single Source Shortest Path, All Pair Shortest Path, Minimum Spanning Tree, etc. run in polynomial time

Intractable Problem:

- A problem that cannot be solved by a polynomial-time algorithm.
- Exponential algorithm will be used to solve the problem.
- The lower bound is exponential.

Exponential Algorithm:

- ➢ An algorithm whose order-of-magnitude time performance is not bounded from above by a polynomial function of n.

Exponential functions:

- ✓ A function of the form k^n is genuinely exponential

  E.g. $O(2^n)$, O(n!), $O(n^n)$

Examples

- ❖ Travelling salesperson, optimal graph coloring, Hamiltonian cycles etc.